

BITWISE OPERATORS

Bitwise operator's works on bits.

Turbo-c is a 16 bit compiler. Due to this bitwise operations are limited to 16 bits only [2^0 to 2^{15}].

Bitwise operators operate **integer** type values only.

We have to calculate only the **on** bits [**1**].

When the first bit[**Sign bit**] is **1** then the number is **Negative** and it is **0** then the number is **positive**.

They are very much used in system software development.

Note: Bitwise operator is low level feature.

C-Language supports following bitwise operators.

& -Bitwise and

| - Bitwise or

^ - XOR ==> Exclusive OR

~ - Compliment operator

<< - Left shift operator

>> - Right shift operator

& - Bitwise and: In this both bits are 1's then result bit is 1. Otherwise result bit is 0.

Eg: **25 & 15 = 9**

25 = 0000 0000 0001 1001
15 = 0000 0000 0000 1111

2 | 25
2 | 12 - 1
2 | 6 - 0
2 | 3 - 0
1 - 1

2 | 15
2 | 7 - 1
2 | 3 - 1
1 - 1

25 & 15 = 9

25 = 0000 0000 0001 1001
15 = 0000 0000 0000 1111

&

0000 0000 0000 1001

↓ ↓
 $2^3 + 2^0$
↓ ↓
8 + 1 = 9

| - Bitwise or: In this both bits are 0's then result bit is 0. Otherwise result bit is 1.

Eg: $25 \mid 15 = 31$

$$25 \mid 15 = 31$$

25 = 0000 0000 0001 1001

15 = 0000 0000 0000 1111

0000 0000 0001 1111

$2^4 + 2^3 + 2^2 + 2^1 + 2^0$

$16 + 8 + 4 + 2 + 1 = 31$

^ - XOR [Exclusive OR]: In this both bits are same then result bit is 0. Otherwise result bit is 1.

Eg: $25 \wedge 15 = 22$

$$25 \wedge 15 = 22$$

25 = 0000 0000 0001 1001

15 = 0000 0000 0000 1111

$\wedge$

0000 0000 0001 0110

$2^4 + 2^2 + 2^1$
 $16 + 4 + 2 = 22$

~ - Compliment operator: In compliment operation the bits are complimented. i.e. 1's become 0's and 0's become 1's. Due to this +Ve no becomes -Ve and -Ve no becomes +Ve.

eg: ~25 → -26

$$\begin{array}{l} 25 = \begin{array}{|c|c|c|c|} \hline 0000 & 0000 & 0001 & 1001 \\ \hline 1111 & 1111 & 1110 & 0110 \\ \hline \end{array} \\ \qquad \qquad \qquad \begin{array}{c} \swarrow \quad \searrow \quad \quad \quad \downarrow \quad \searrow \\ -128 + 64 + 32 + 4 + 2 = -26 \\ -128 + 102 = -26 \end{array} \end{array}$$

$$25 = 0000\ 0000\ 0001\ 1001$$

$$\sim = 1111\ 1111\ 1110\ 0110$$

$$\begin{array}{c} \backslash \quad \backslash \\ 5 \quad 2 \end{array}$$

$$2+4+32+64+128+256+512+1024+2048+4096+8192+16384-32768=-26$$

$$\sim -25 = 0000\ 0000\ 0001\ 1001$$

$$1's\ \sim = 1111\ 1111\ 1110\ 0110$$

$$2's\ \sim = \begin{array}{r} 0000\ 0000\ 0000\ 0001 \\ 1111\ 1111\ 1110\ 0111 \end{array}$$

$$\begin{array}{c} \swarrow \quad \searrow \\ 24 \quad 2 \\ 16+8=24 \end{array}$$

$$\begin{array}{cccc} 1 & 0 & 0 & 01 \\ 0 & 1 & 0 & 1 \\ \hline 1 & 1 & 0 & 10 \end{array}$$

Note: When starting bit is 1 given no is –Ve.

Eg: $\sim -25 \rightarrow +24$

$$\sim -25 = +24$$

$$25 = \begin{array}{|c|c|} \hline 0000 & 0000 \\ \hline \end{array} \quad 0001\ 1001$$

$$\begin{array}{|c|c|} \hline 1111 & 1111 \\ \hline \end{array} \quad 1110\ 0110$$

<== 1's compliment

$$+1$$

<== 2's Complement

$$\begin{array}{r} \sim \\ \hline 1111\ 1111\ 1110\ 0111 \\ 0000\ 0000\ 0001\ 1000 \end{array}$$

$$\downarrow \quad \downarrow$$

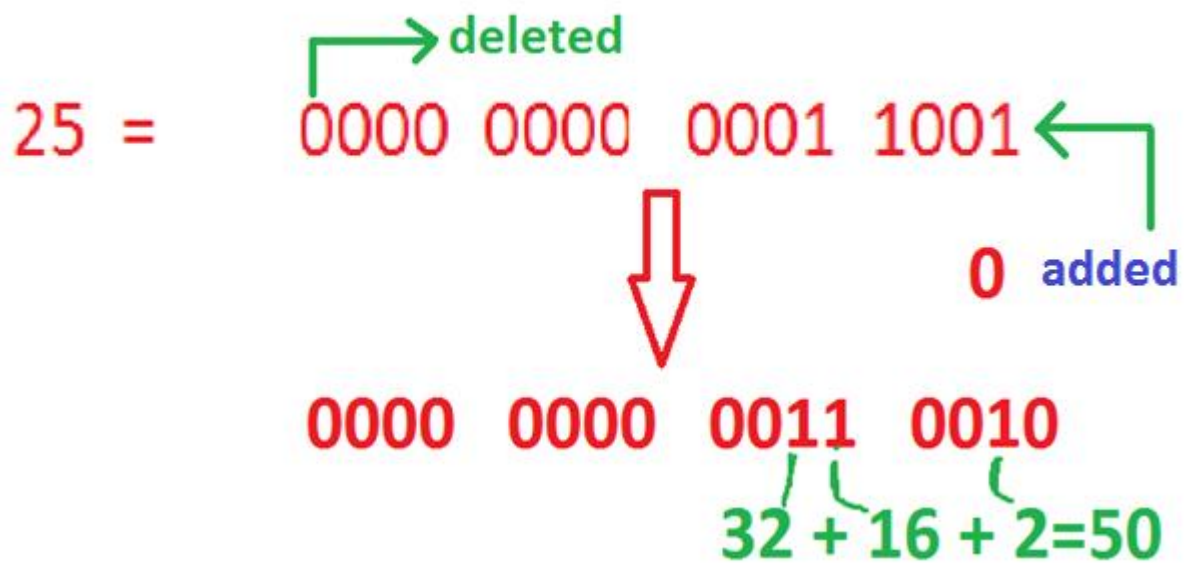
$$16+8=24$$

<< - left shift operator:

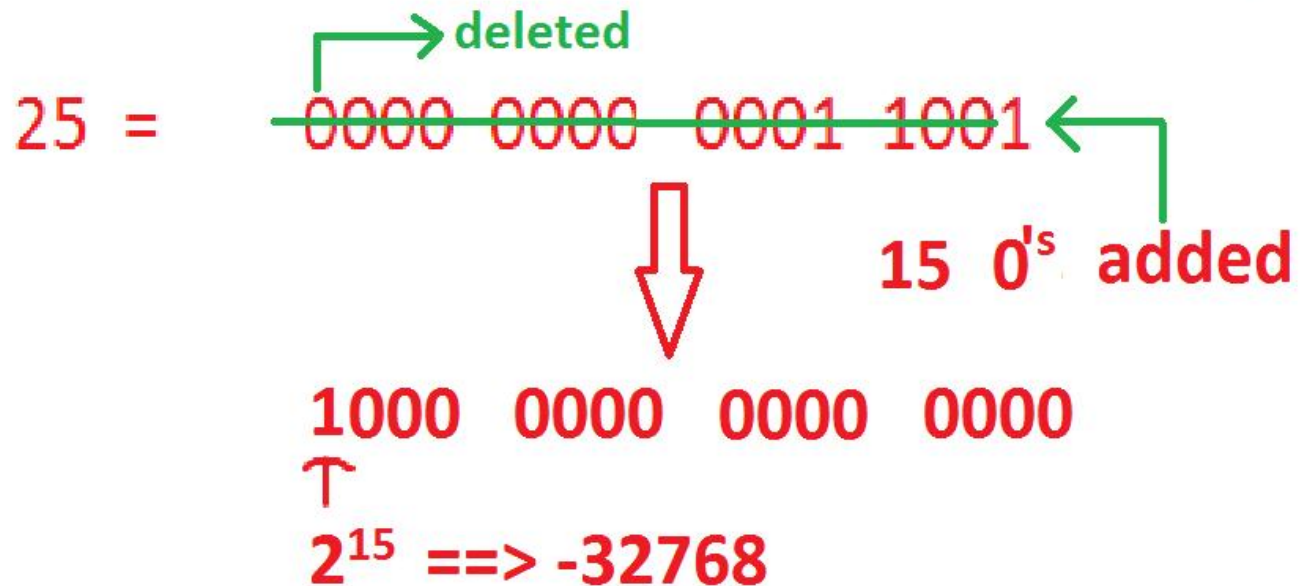
In left shift operation, the specified no of bits are deleted from left side and the same no of **zeros** added on right side. In left shift operation, most probably the value is multiplied with 2 that no of times.

Eg: $25 \ll 1 = 50$, $25 \ll 2 = 100$, $25 \ll 15 = -32768$,
 $25 \ll 16 = 0$

eg: $25 \ll 1 = 50$



eg: $25 \ll 15 = -32768$



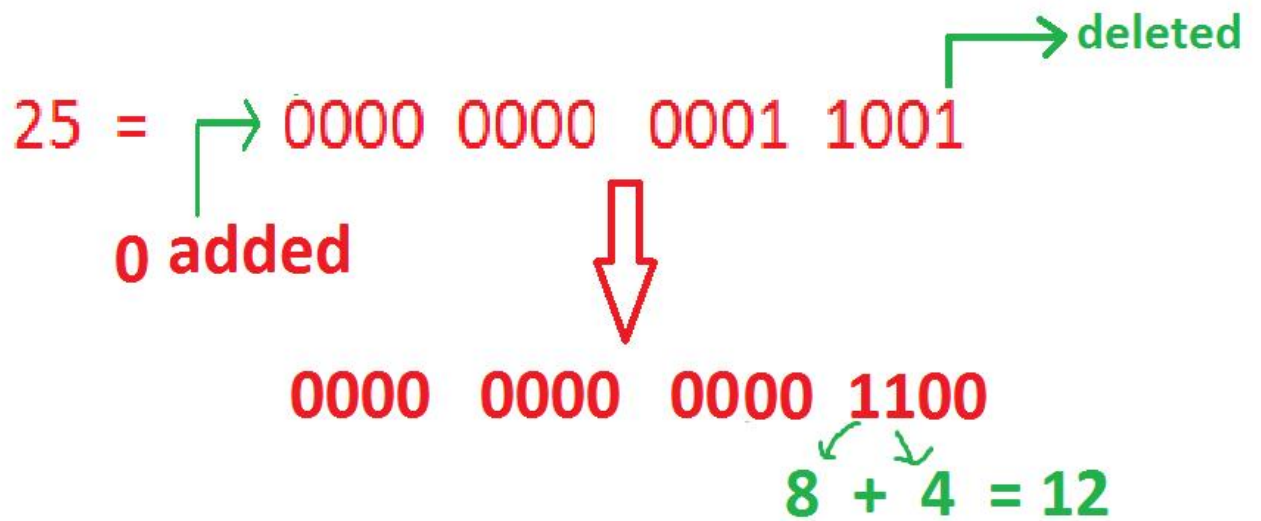
Note: When starting bit 1 no is negative.

>> - Right shift operator:

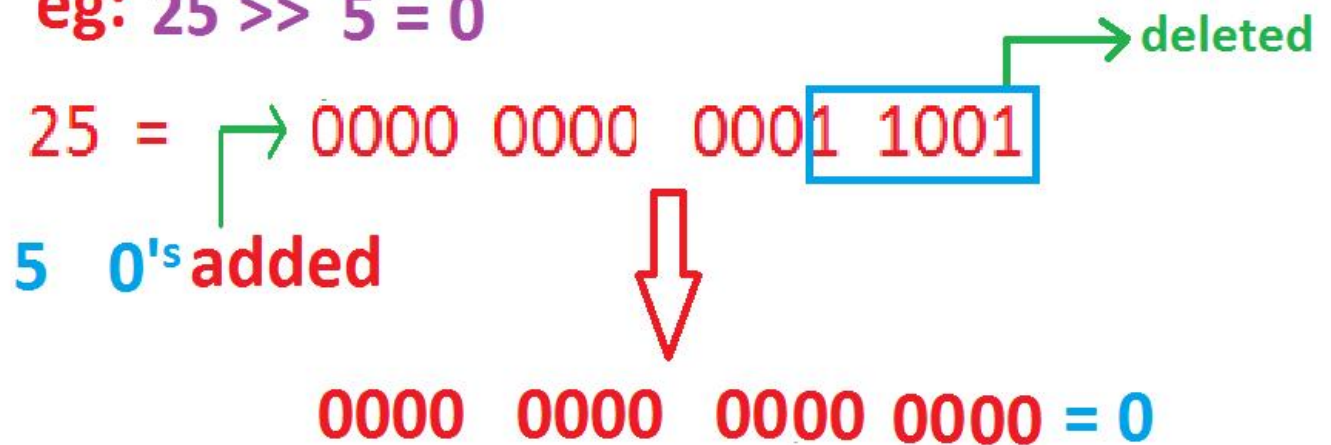
In right shift operation, the bits are moved to right side i.e. the specified no.of bits are deleted from right side and same no.of **zero's** are added left side. Due to this always the number is divided with 2 that no of times.

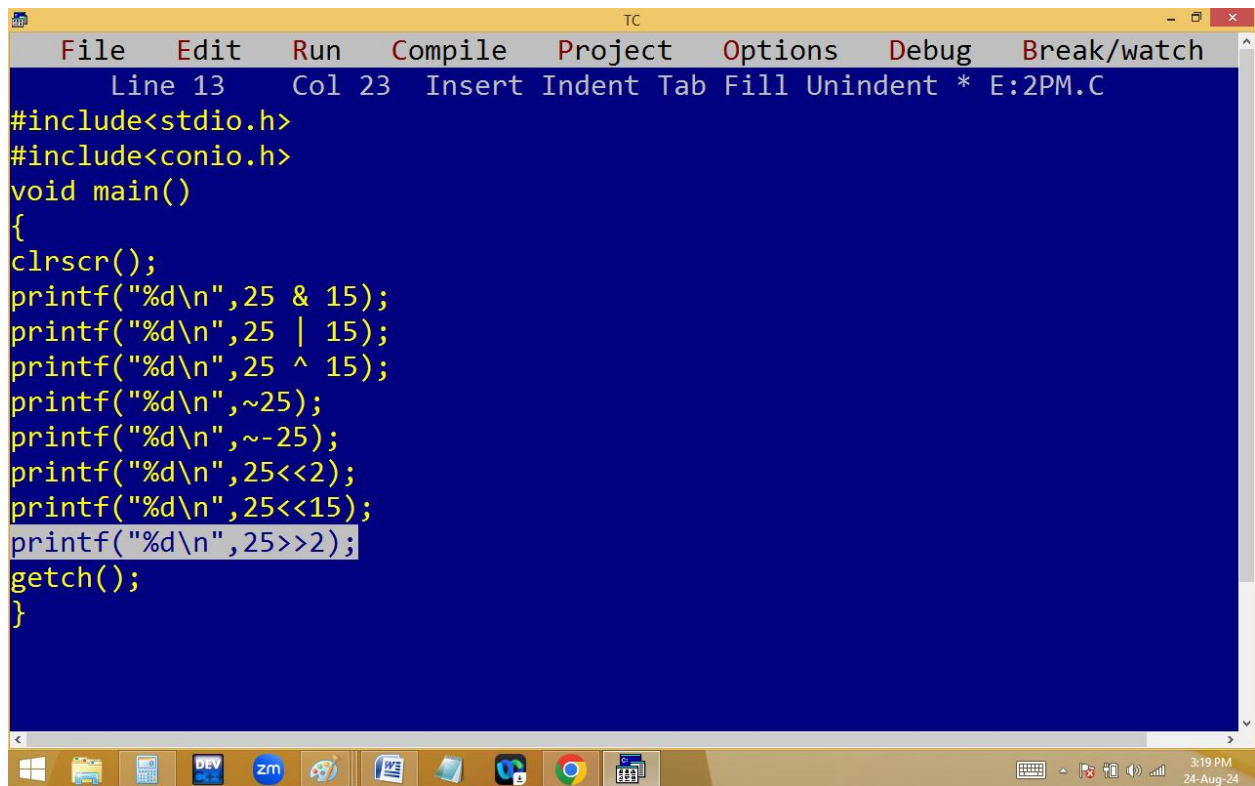
Eg: $25 \gg 1 = 12$, $25 \gg 2 = 6$, $25 \gg 3 = 3$, $25 \gg 4 = 1$, $25 \gg 5 = 0$

eg: $25 \gg 1 = 12$



eg: $25 \gg 5 = 0$

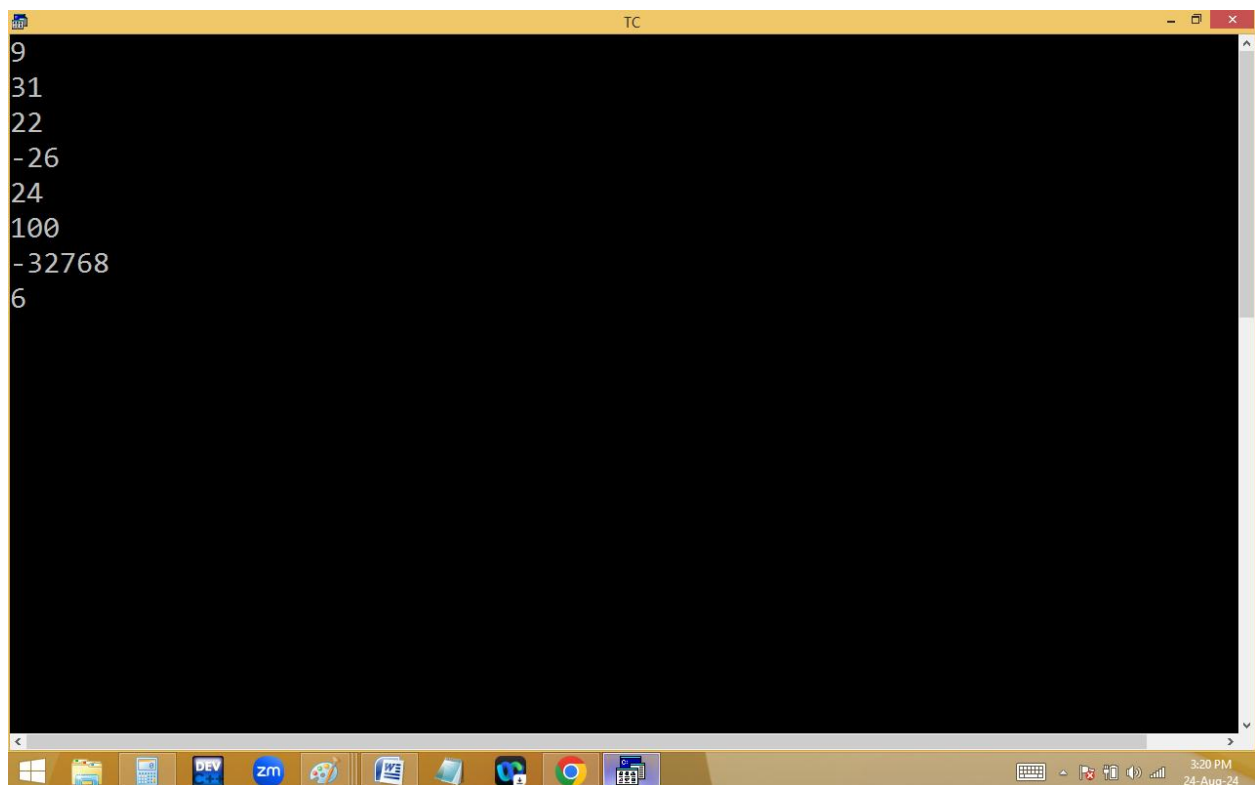




The screenshot shows the Turbo C++ (TC) IDE with a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch) and a status bar (Line 13, Col 23, Insert, Indent, Tab, Fill, Unindent, * E:2PM.C). The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",25 & 15);
printf("%d\n",25 | 15);
printf("%d\n",25 ^ 15);
printf("%d\n",~25);
printf("%d\n",~-25);
printf("%d\n",25<<2);
printf("%d\n",25<<15);
printf("%d\n",25>>2);
getch();
}
```

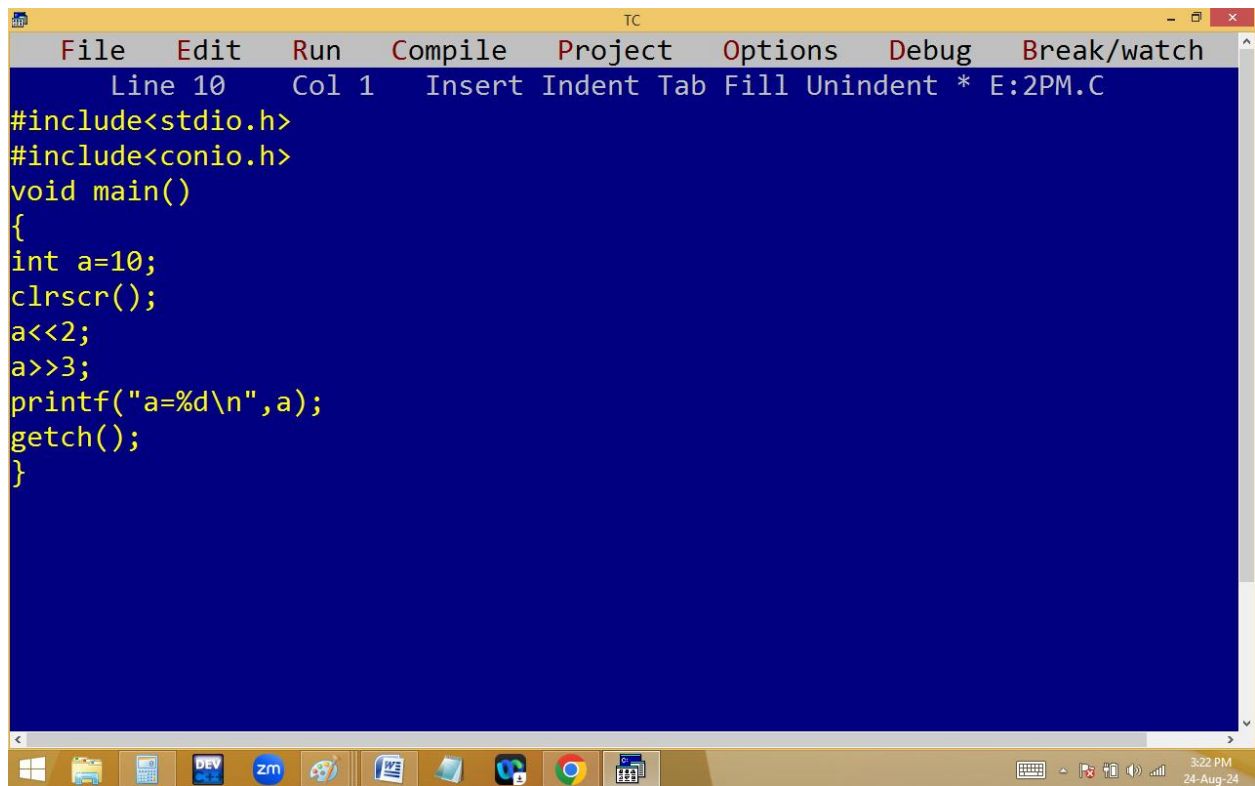
The taskbar at the bottom shows various application icons and the system clock indicating 3:19 PM on 24-Aug-24.



The screenshot shows the Turbo C++ (TC) IDE with the same menu bar and status bar. The output window displays the results of the bitwise operations performed on the number 25:

```
9
31
22
-26
24
100
-32768
6
```

The taskbar at the bottom shows the same application icons and the system clock indicating 3:20 PM on 24-Aug-24.



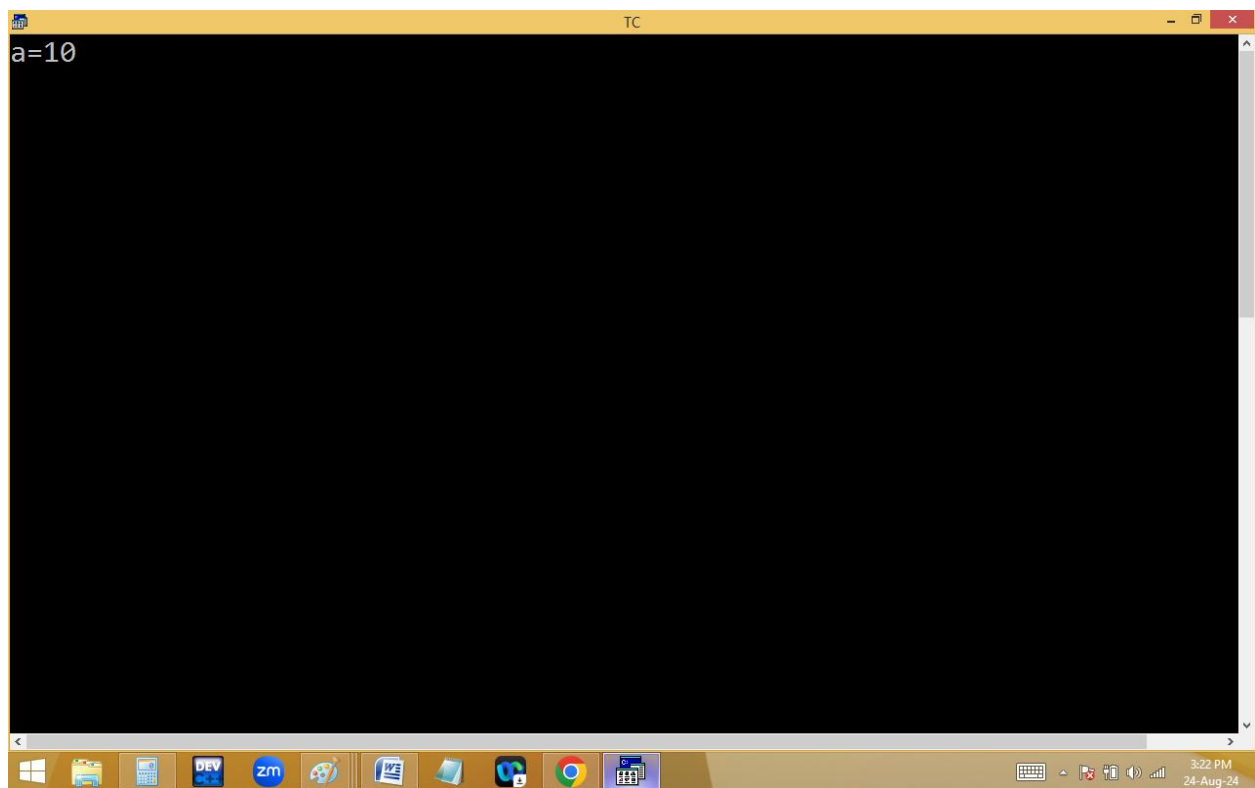
TC

File Edit Run Compile Project Options Debug Break/watch

Line 10 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
a<<2;
a>>3;
printf("a=%d\n",a);
getch();
}
```

3:22 PM 24-Aug-24



TC

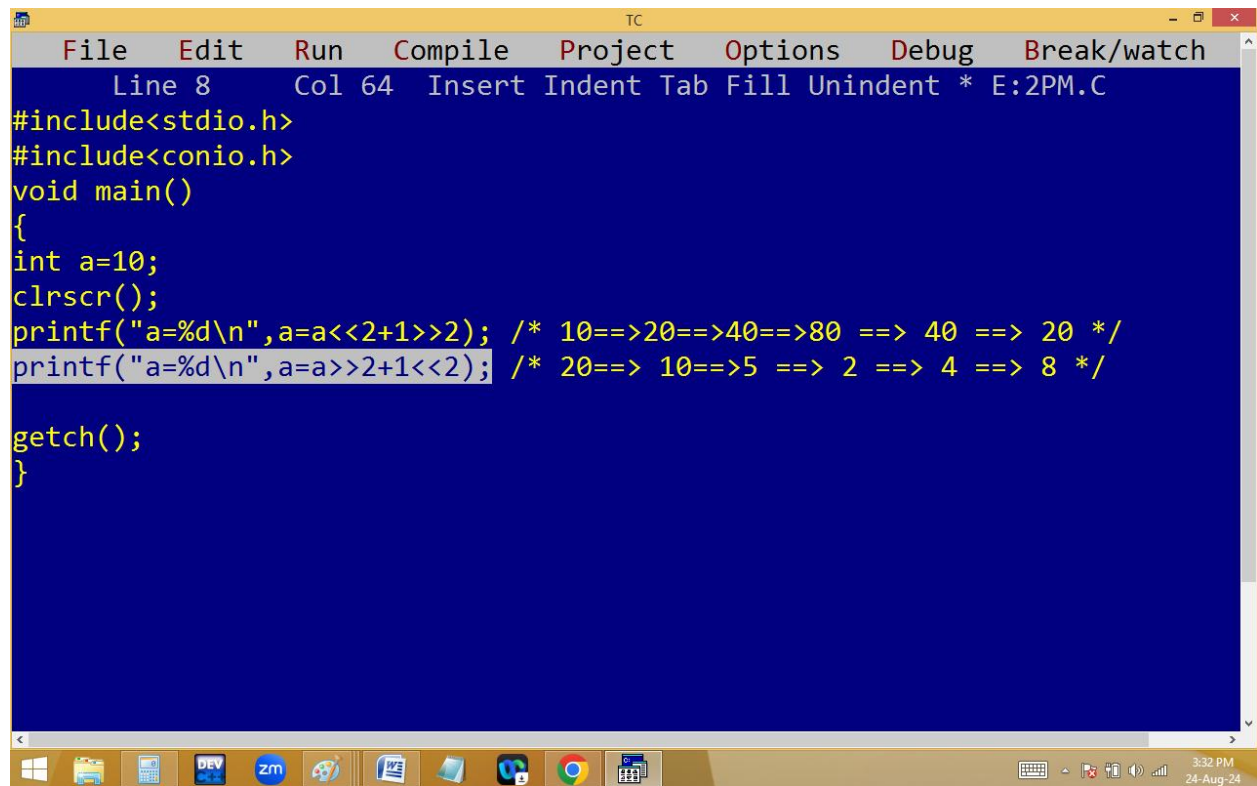
```
a=10
```

3:22 PM 24-Aug-24

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 35 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
a<<=2; /* a=a<<2; 10==>20==>40 */
a>>=3; /* a=a>>3; 40==>20==>10==>5 */
printf("a=%d\n",a);
getch();
}
```

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 65 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
printf("a=%d\n",a<<2+1>>2); /* 10==>20==>40==>80 ==> 40 ==> 20 */
printf("a=%d\n",a>>2+1<<2); /* 10==>5 ==> 2 ==> 1 ==> 2 ==> 4 */

getch();
}
```



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". The status bar at the top indicates "Line 8", "Col 64", and "Insert Indent Tab Fill Unindent * E:2PM.C". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
printf("a=%d\n",a=a<<2+1>>2); /* 10==>20==>40==>80 ==> 40 ==> 20 */
printf("a=%d\n",a=a>>2+1<<2); /* 20==> 10==>5 ==> 2 ==> 4 ==> 8 */

getch();
}
```

The second printf statement and its comment are highlighted with a light blue background. The Windows taskbar is visible at the bottom, showing icons for various applications and the system clock indicating 3:32 PM on 24-Aug-24.